

Markscheme

Mathematics: analysis and approaches

Practice question bank

42. (a)

Given that $\ln A = 2 \ln x + 3 \ln y - \ln z$, where $x = 3$, $y = 2$ and $z = 6$.

Find the value of A .

$$\ln A = \ln \left(\frac{x^2 y^3}{z} \right) \Rightarrow A = \frac{x^2 y^3}{z} \quad (M1)$$

$$A = \frac{(3)^2 (2)^3}{6} = \frac{72}{6} \quad (M1)$$

$$A = 12 \quad \mathbf{A1}$$

[3 marks]